
LESSON 1.3: Route Channel Template

What this lesson is about

This lesson describes about Sterling File Gateway Routing Channel Templates (RCT) and the file structures.

What you should be able to do

- Describe the components of a Routing Channel Template.
- Explain the difference between System facts and Custom facts.
- Describe the file structures usage in Sterling File Gateway.
- Explain the difference between container and non-container file layers.

How you will check your progress

The progress of the course is analyzed based on the successful application of the hands-on exercises.

Additional References

Documentation

- Routing Channel Template:
http://www.ibm.com/support/knowledgecenter/SS4TGX_2.2.0/com.ibm.help.sfg_int_arch.doc/sfg_int_arch_plugin-gentopic3.html?lang=en
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Route channel templates

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Introduction

The routing channel template (RCT) is a fundamental concept in the Sterling File Gateway application:

- The Routing Channel Template defines the structure through which routing occurs.
- The Routing Channel Template specifies consumer and producer mailbox structures and file structures. It functions as a policy that mandates which partners can participate in various file transfer scenarios.
- A Routing Channel Template is required to create a routing channel, which establishes the producer-consumer relationship for file transfers.

Creating route channel template

Routing channel templates are created by users that are defined with Integration Architect access.

Routing channel templates consists parameters:

- Template type - static or dynamic
- Special characters
- Groups
- Provisioning Facts
- Producer mailbox and file structure
- Consumer delivery channel

The screen shot shows the types of routing channel templates

Routes Participants Tools Profile

Template Name :

Type Special Characters Groups Provisioning Facts Producer Consumer

Template Type
Select the template type to indicate how consumers are identified.

☒ Static - consumer is selected when channel created

☐ Dynamic - consumer is determined at time of routing

Route channel templates

Creating route channel template

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Template type

Routing channel templates can be *static* or *dynamic*.

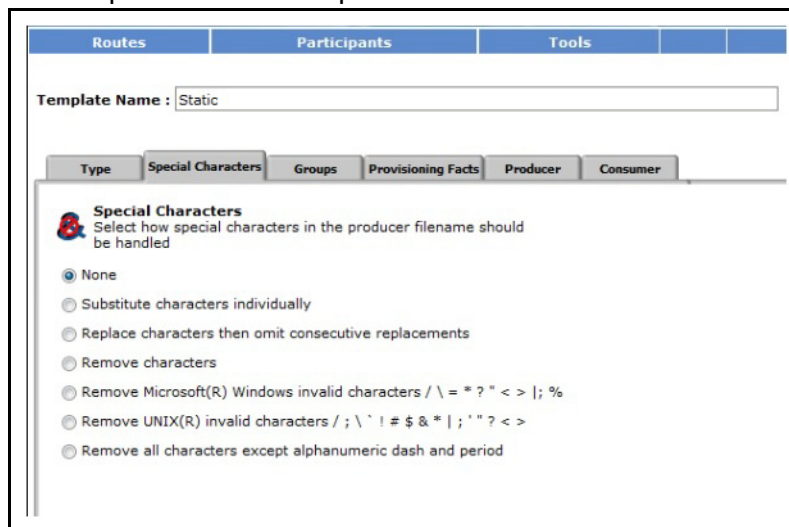
- A static routing channel template explicitly defines the consumer.
- A dynamic routing channel template does not require that the consumer be explicitly specified when the routing channel is created.

A dynamic routing channel template can be used to determine the consumer at the time of file transfer.

Special Characters

Sterling File Gateway provides various options for handling special characters in file names. This option provides flexibility when the file name provided by the producer is not compatible with the consumer file naming convention.

The screen shot shows the options to handle special characters in file names.



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Route channel templates

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Creating route channel template

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Special Characters

The table describes the special character handling options:

Selection	Results
None	Do not change any special characters.
Substitute characters individually	Enter the new characters to be substituted for the previously existing characters. The number of characters that are specified in the search for and replace fields must match. The characters are case-sensitive. For example, Search for character sequence ABC, Replace with 123 would translate a producer file name of: ■ Abbot.txt to a consumer file name of 1bbot.txt ■ ABBOT.TXT to a consumer file name of 122OT.TXT Search for abcdef, Replace with ABCDEF would translate producer a file name of: ■ abbot.txt to a consumer file name of ABBot.txt
Replace characters, then omit consecutive replacements	Enter which sequence to substitute with what new sequence. For example, Search for [*%\$] Replace with -, would translate all of the following producer file names: ■ A*B.txt ■ A***B.txt ■ A%%%%B.txt ■ A*B.txt ■ A\$*\$*%B.txt ■ A*\$%\$*\$*%B.txt
Remove characters	Enter the characters to remove from the producer file name to create the consumer file name.
Remove Windows invalid characters \:*?"<> ;,%	Characters that are invalid or cause problems in the Windows operating system are removed from producer file names before delivering to consumers.
Remove UNIX invalid characters /;!#\$%&* '?"<>)	Characters that are invalid or cause problems in the UNIX operating system are removed from producer file names before delivering to consumers.
Remove all characters, except alphanumeric, dash, and period	All special characters are removed from producer file names before delivering to consumers.

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Route channel templates

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Creating route channel template

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Groups

The Groups tab allows you to select the producer groups and consumer groups that can use the template. Only members of these groups are eligible partners for the routing channel.

The screen shot displays the Groups tab of the route channel template.

The screenshot shows the 'Groups' tab of a route channel template configuration. At the top, there is a navigation bar with tabs: Routes, Participants, Tools, Profile, and Help. Below this, a 'Template Name' field is set to 'Static'. The 'Groups' tab is selected, showing a sub-tab bar with 'Type', 'Special Characters', 'Groups', 'Provisioning Facts', 'Producer', and 'Consumer'. The 'Groups' sub-tab is active, displaying the instruction 'Identify groups eligible to use this template'. There are two main sections: 'Producer Groups' and 'Consumer Groups'. Both sections have a list box containing 'SFG'. At the bottom of each list box are 'Add' and 'Delete' buttons.

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Route channel templates

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Creating route channel template

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Provisioning facts

Provisioning facts provide a controlled way to customize particular routing channels within the context of a particular routing channel template.



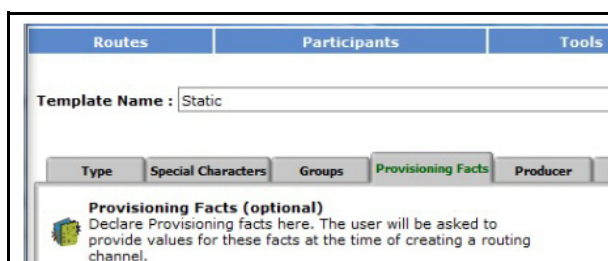
Example

To create a mailbox of the form /PartnerName/UserId/

The following are some of the key attributes of provisioning facts:

- The value of user ID is bound to the new routing channel and can be used for creating mailboxes (at either provision time or routing time) and for creating file names (at routing time).
- Provisioning facts are available for static and dynamic templates.
- You are not required to define any provisioning facts when creating a routing channel template.
- You can edit provisioning facts as part of routing channel template edit. However, provisioning fact values cannot be edited after being set in a routing channel.
- You can add a maximum of 10 provisioning facts to a routing channel template.
- The same provisioning fact can be used in more than one template. For example, both template 1 and template 2 can have a "myUserID" provisioning fact.
- Provisioning Facts are immutable and once they are set during Routing Channel
- Provisioning facts can be mutated in a custom business process that will have visibility to the provisioning facts.
- The four properties included in the filegateway.properties represents as:
 - 1 property in custom business process that will get visibility and can change them during routing.
 - 2 properties which will identity the consumer mailbox prefix
 - Another property business process can be optionally invoked in FileGatewaySendMessage BP to extract data before delivery.

The screen displays the Provisioning Facts tab of a routing channel template.



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Route channel templates

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Creating route channel template

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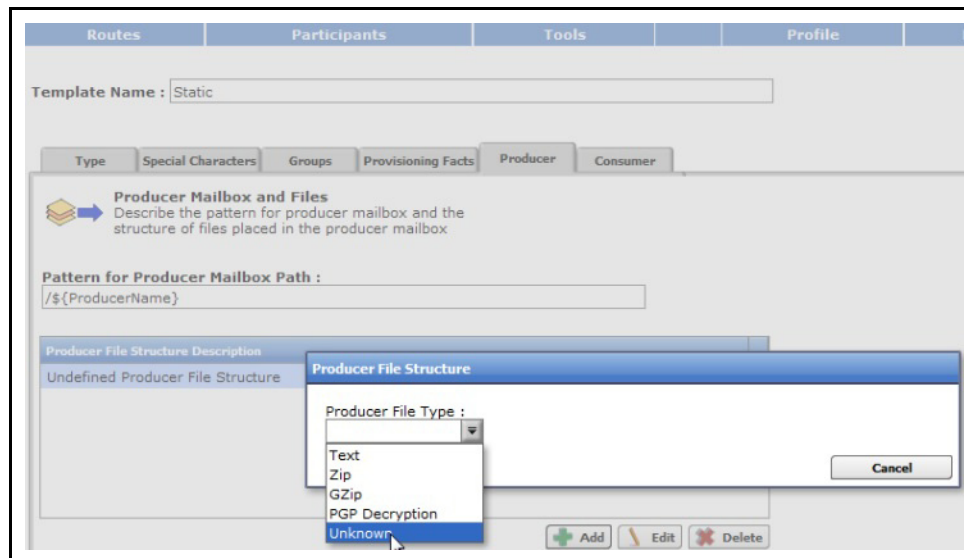
Producer

The producer tab allows you to specify the producer mailbox name pattern and producer file structure. Mailbox paths are **not** case-sensitive and should be unique without regard to case; mailboxes can be created by the system. The mailbox is used to define the location of the outgoing files in the producers mailbox structure.

The Producer File Structure is consisted of the following information:

- File type
- File name patter as regular expression
- File name pattern group fact names, comma delimited

The screen shot displays the selection of the Producer File Type in the Producer tab of the routing channel template.



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Route channel templates

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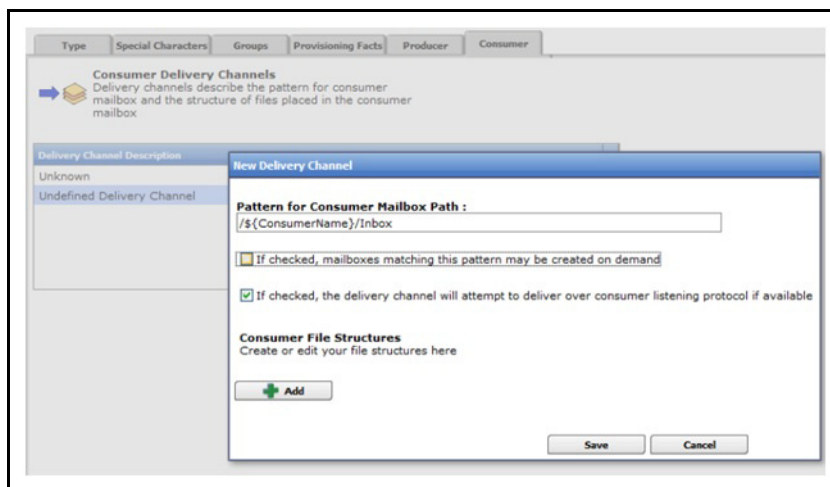
Creating route channel template

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Consumer

The consumer tab allows you to specify the delivery channel, including the consumer mailbox pattern and consumer file structure. You can check whether you would like Sterling File Gateway to create mailboxes on demand as needed.

The following image displays the setting up of a new delivery channel in the Consumer tab of routing channel template.



Facts

Introduction

Facts are name-value-pairs that contain information about active routes and the files that are being routed.

Some facts are intrinsic to the route, where others are determined during processing of the producer file structure. A fact is a named piece of known information about an executing routing channel, for example:

- A part of a file name
- The time a message arrived from the producers transfer channel
- The name of the producer
- The name of the consumer

A known fact is frozen. Facts never lose their value or change values.

Facts Type

The types of facts in Sterling File Gateway are:

- System facts
- Custom facts

System facts

Sterling File Gateway provides system facts and supports the specification of custom facts during the creation of routing channel template. Most system facts can be used to build out the pattern for producer and consumer mailboxes and to define the producer and consumer file structures in the routing channel template.

For example,

- For the producer, as you are defining the variable, such as, the first section of a file name is, the ConsumerName.
- For the consumer, you can use the system facts to control what the end file name of the received file, based on what the producer sent.

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Facts

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Introduction

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The following table lists the System facts:

Fact Name	Description
ConsumerBroadcastGroup	The name of the consumer group to which files must be broadcast.
ConsumerName	The name of the consumer.
ConsumerCode	The code of the consumer.
ProducerFilename	The name of the file that the producer sent.
ProducerName	The name of the producing partner.
ProducerCode	The code of the producing partner.
RoutingTimestamp	The time when the routing began (when the route was identified).
ConsumerPgpExtension	The extension of a file that is based on the consumer's PGP preferences. Valid values are: ■ .pgp - consumer is getting a PGP file ■ .asc - consumer is getting an ascii encoded pgp file ■ (nothing) - consumer is not getting a PGP file Only available for the PGP file layer type.

When using system facts in the producer and consumer mailbox or in the consumer file structure, following notations are required:

Notation	Description
<code>\${factname}</code>	Replaced with the value of factname.
<code>\${fmt:factname}</code>	Replaced with the value of factname, formatted according to fmt. The fmt string can be any format string that is supported by java.util.Formatter or a Timestamp value.
<code>\${x:y:z}</code> (any use of :)	Any use of more than one colon is an error and is rejected at Routing Channel Template creation time.
<code>\$\$ {</code>	Produces a literal \$ {
<code>{x:y:z}</code>	Any use of more than one colon is an error and is rejected at Routing Channel Template creation time.
<code>{factname</code>	An unclosed reference is an error and is rejected.
<code>{factname}</code>	Is not an error, but produces a literal {factname} and is not replaced.

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Facts

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Introduction

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Custom facts

User defined facts are called custom facts. The Sterling File Gateway states rules for creating customer facts. User-created facts must be prefixed with the string **my** to clearly delineate between custom and system facts. Custom facts are used in producer file structure, consumer mailbox pattern and in the consumer file structure. For example,

- myFileName
- myDateTime
- myConcatenatedFilename.

Example

Consider the following example. The file name format pattern uses both system and custom facts:

`${mySanitizedFilename}_${RoutingTimestamp}.zip`

In the example, *mySanitizedFilename* was defined as a custom fact in the producer file structure, and *RoutingTimestamp* is a system fact. When Sterling File Gateway receives the file sent by producer, it derives the facts and then uses the facts to determine the values to place in the consumer file name.

Example formats and results are:

- No format - For most facts, this results in their value. For *RoutingTimestamp*, this format will produce an eight digit string, such as, "20100821" for August 21, 2010.
- s - The value of the fact
- S - The value of the fact that is mapped to all uppercase
- .5s - the value of the fact that is right truncated to five characters

Routing timestamp

RoutingTimestamp uses Java Formatter class formats with an enhancement that allows multiple formats to be specified for one value. Essentially, this value is either a 't' or 'T' followed by multiple time format specifiers. 'T' forces the results to uppercase.

Examples of *RoutingTimestamp* are:

- tYmd - An 8-digit string consisting of a 4-digit year, 2-digit month, and 2-digit day of the month, such as, 20080201 (default).
- tHMS - A 6-digit Hour, Minute, Second value using a 24 hour clock, such as, 053027.
- tIMSp - A 5 or 6-digit Hour, Minutes, Seconds value using a 12 hour clock, such as, 53027am.
- TIMS' 'p - Similar to the previous one, but in uppercase with an embedded space, such as, 53027 AM.
- Ta'-b'-d'-Y - Textual date, in uppercase., such as, FRI-AUG-21-2008.

File Structure

Producer file structure

A producer file structure contains the basic content structure description of a file and its naming conventions.

The Integration Architect specifies a pattern for the producer mailbox and a producer file structure when creating a routing channel template.

The default pattern for the producer mailbox path is:

`/${ProducerName}`

You can change it using regular expressions and facts. Only the following facts are valid in the pattern for the producer mailbox path:

- `ProducerName`
- `ProducerCode`
- `ConsumerName`
- `ConsumerCode`



Note

If creating a dynamic routing channel template, do not specify `ConsumerName` or `ConsumerCode` in the producer mailbox path.

A producer file structure specifies the type of file that producers are expected to send to Sterling File Gateway. A file structure is an ordered list of file layers. A producer file structure can have one or more file layers. The layers are of a certain type, have a specific regular expression pattern, and can specify facts to be defined when a file matching the pattern is received.

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File Structure

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File structure layers

A file structure is an ordered list of file layers from outermost file structure to innermost file structure. A producer file structure can have one or more file layers. The file structure layers have a specific regular expression patterns, and specify facts to be defined when a file matching the pattern is received.

There are two types of layers:

- Container layer
- Non-container layer (primitive)

Container

Container contains other container or non-container layers. For example, a PGP layer can contain a compressed file layer and a compressed file layer can contain a TEXT layer. A compressed file can contain multiple files only, if it is the outermost layer. All layers except the innermost layer must be one of the following compressed file

- Gzip
- PGP

Non-container

Non-container does not contain any nested layers. The innermost layer must be a primitive layer of type:

- Text
- Unknown

There is no limit to the number of layers for a file structure. Certain processing operations, such as the special character handling feature, apply to the outermost layer only. Also, route identification occurs at the outermost layer, except in the case of a compressed file layer. With a compressed layer, the second layer is also used to perform route identification.

If the layer type of the consumer file structure is text, the layer types of every producer file structure must be text and must not be unknown. If the layer type of the consumer file structure is unknown, the producer file structure can include either unknown or text.

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File Structure

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Consumer file structure

A consumer file structure contains the basic content structure description of a file and its naming conventions.

A consumer file structure specifies the type of file that consumers expect from Sterling File Gateway. A file structure is an ordered list of file layers, from outermost file structure to innermost file structure. The layers are of a certain type and they can have a specific format pattern and can include unique parameters.

The Integration Architect must create a consumer file structure when creating a delivery channel.

To create a file structure with multiple file layers, the outside layer must be a container layer of type:

- Compressed file
- Gzip
- PGP

The innermost layer of a producer file structure must be a non-container layer of type:

- TEXT
- Unknown

If the layer type of the consumer file structure is text, the layer types of every producer file structure must be text and must not be unknown. If the layer type of the consumer file structure is unknown, the producer file structure can include either unknown or text.

For each layer, the file name format pattern is used to generate the new file name. The file name format pattern uses system and custom facts to define how the consumer file should be named.

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File Structure

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Regular expression

Regular expression is a powerful standardized pattern matching language. The Sterling File Gateway application can match file names and gather facts that are related to file names.

A high level, sample regular expression is shown in the image.

Regular Expressions - Quick Reference Guide		
Anchor \$ start of line \b end of line \B word boundary \A not at word boundary \a start of subject \g first match in subject \z end of subject \Z or before newline at end	Literal Characters Letters and digits match exactly a x B 7 0 Some special characters match exactly @ - = % Escape other specials with backslash \ \ \$ \%	Character group contents x individual chars x-y character range [:class:] posix char class [^class:] negated class
Non-printing characters \a alarm (BEL, hex 07) \b "control-" \e escape (hex 1B) \f formfeed (hex 0C) \n newline (hex 0A) \r carriage return (hex 0D) \t tab (hex 09) \ddd octal code dddd \xhh hex code hh \x(hhh..) hex code hhhh..	Character Groups Almost any character (usually not newline) [.] Lists and ranges of characters [a-z] Any character except those listed [^a-z]	Examples [a-zA-Z0-9_] [[:alnum:]]
Generic character types \d decimal digit \D not a decimal digit \s whitespace character \S not a whitespace char \w "word" character \W "non-word" character	Counts (add ? for non-greedy) 0 or more ("perhaps some") [*] 0 or 1 ("perhaps a") [?] 1 or more ("some") [+] Between "n" and "m" of [n,m] Exactly "n", "n" or more [(n), (n,)]	Comments (?#comment) Conditional subpatterns (?condition)yes-pattern (?condition)yes no-pattern
POSIX character classes \alnum letters and digits \alpha letters \ascii character codes 0-127 \blank space or tab only \cntrl control characters \digit decimal digits \graph printing chars -space \lower lower case letters \print printing chars +space \punct printing chars -alnum \space white space \upper upper case letters \word "word" characters \xdigit hexadecimal digits	Alternation Either/or [a b] Lookahead and Lookbehind Followed by [?(=)] NOT followed by [?(!)] Following [?(<=)] NOT following [?(<!)]	Recursive patterns (?n) Numbered (?0) (?R) Entire regex (?&name) Named Replacements \$n reference capture Case foldings \u upper case next char \U upper case following \l lower case next char \L lower case following \E end case folding
	Grouping For capture and counts ([...]) Non-capturing (?:...) Named captures (?<name>...) Back references Numbered \n \gn \g{n} Relative \g{-n} Named \k<name>	Conditional insertions (?n:insertion) (?n:insertion;otherwise)

Some of the examples with description of regular expression are shown in the table.

Pattern	Description
.+	Simplest pattern that matches any name.
@, %, =	A literal character represents special characters that matches exactly.
\L	A lower case folding of the string or character.
\s or \S	A generic character representing whitespace or (\S) not a whitespace character.

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File Structure

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Regular expression

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Facts are useful for describing the meaning of the regular expression language.

The regular expression carries in it the definition of the file that is expected from the producer. Because a file name can carry different categories of information, it is helpful to chunk those categories of information into groups.

In the regular expression, each group of information is contained within parentheses "()". Each group within the regular expression must have a corresponding fact, either system or custom.

You can use the same fact in different layers of a producer file structure or in multiple file structures for the same routing channel template.

The value of this fact must not change during execution. If it does change, Sterling File Gateway generates an error.

The simplest pattern that matches any name, and defines no facts, is: .+

The File name pattern as a regular expression field is not case-sensitive when matching file names.

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File Structure

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Regular expression example

Consider the following example of file format in Regular Expression

(\p{Alnum}+[-_]?(\d{4})_(\d{2})_(\d{2})\.zip)where:

- the full expression, (\p{Alnum}+[-_]?(\d{4})_(\d{2})_(\d{2})\.zip), = capturing group 1 with fact = "mySanitizedFilename"
- (\d{4}) = capturing group 2 with fact = "myFileYear"
- (\d{2}) = capturing group 3 with fact = "myFileMonth"
- (\d{2}) = capturing group 4 with fact = "myFileDay"

The corresponding file name pattern group fact name entry for this expression is:

mySanitizedFilename,myFileYear,myFileMonth,myFileDay

Applying this file format to the file: PO_2008_08_20.zip

Will give the following results:

- mySanitizedFilename= PO_2008_08_20.zip
- myFileYear=2008
- myFileMonth=08
- myFileDay=20

Consumer side regular expression example

In this example, the file name format pattern uses both system and custom facts:

\${mySanitizedFilename}_\${RoutingTimestamp}.zip

mySanitizedFilename was defined as a custom fact in the producer file structure, and *RoutingTimestamp* is a system fact. When Sterling File Gateway receives the file from producer, it derives the facts and then uses the facts to determine the values to place in the consumer file name.

Exercise 1.3.1: Create a Routing Channel Template (RCT)

Introduction

In this exercise, you will create a static route channel template that finds a file in the producer's mailbox of /partner and routes the data to the consumer's mailbox of /abc/inbox. The static channel is one producer to one consumer (1-1) relationship. The producer uploads any file into the repository from myfilegateway and the RCT routes the data, keeping the original file name.

Instructions

Complete the following steps to create a Route Channel Template:

Step 1: Log in to filegateway by specifying **fg_sysadmin** as the user ID and **password** as password.

Step 2: From the main Menu, select **Routes > Templates**.

Step 3: Click **+Create**.

Step 4: Enter **Static** for the **Template Name**.

Step 5: Select **Static** tab for the **Template Type**.

Step 6: Click the **Special Characters** tab. Select **None** for the **Special Characters**.

Step 7: Click the **Groups** tab. Select **+Add** for the **Producer Groups**.

Step 8: From the drop down list, select **SFG**.

Step 9: Select **+Add** for the **Consumers Groups**.

Step 10: From the drop down list, select **SFG**.

Step 11: Click the **Provisioning Facts** tab. Accept the default for the **Provisioning Facts**.

Step 12: Click the **Producer** tab. Select **+Add** for the **Producer**.

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Exercise 1.3.1: Create a Routing Channel Template (RCT)

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Instruction

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Step 13: From the **Producer File Structure** screen, specify the following values:

Parameter	Value
Producer File Type:	Unknown
File name pattern as regular expression:	.+
File name pattern group fact names, comma delimited:	<leave blank>

Step 14: :Click **Save**.

Step 15: Click the **Consumer** tab.

Step 16: Select **+Add** for the **Consumer**.

Step 17: Accept the default value for mailbox path.

Step 18: Select, **If checked, mailboxes matching this pattern may be created on demand**.

Step 19: Click **+Add** to access the **Consumer File Structure** screen. Specify the following parameters:

Parameter	Value
Consumer File Type:	Unknown
File name format:	\${ProducerFilename}

Step 20: Click **Save** three times.

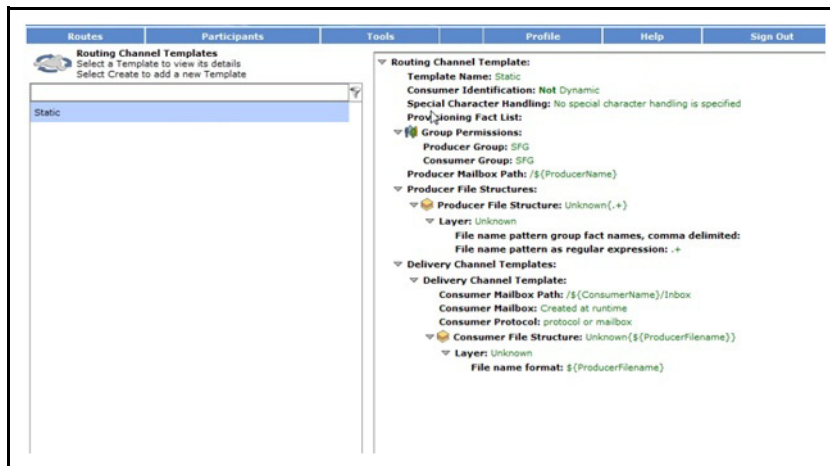
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Exercise 1.3.1: Create a Routing Channel Template (RCT)

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Instructions

Step 21: The routing channel template successfully created message is displayed. Click **OK**. You will see configured RCT details as shown in the screen shot.



Lesson review

What you have been able to do

- Describe the components of a Routing Channel Template.
 - Explain the difference between System facts and Custom facts.
 - Describe the file structures usage in Sterling File Gateway.
 - Explain the difference between container and non-container file layers.
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